

AREKANTI SIDDARTHA

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EDUCATION

SRM University Amaravati, Andhra Pradesh
B.Tech in Computer Science and Engineering; CGPA: 7.57/10.0

Jun 2021 – May 2025

EXPERIENCE

Software Engineering Intern

May 2024 – Jul 2024

SECENAI Semiconductors and Test Solutions Pvt. Ltd., Amaravati, Andhra Pradesh

- Architected and deployed an automated facial recognition attendance system processing 100+ students daily, eliminating 25+ hours/week of manual data entry across the organization
- Achieved 95%+ identification accuracy by training custom CNN models on 10,000+ facial images using transfer learning with VGGFace and FaceNet architectures
- Enhanced real-time detection pipeline using OpenCV and SVM classifiers, reducing average processing time from 3.2s to 0.8s per frame (75% improvement)

PATENTS & PUBLICATIONS

FEELGAN: Federated Edge Learning with GANs for Medical Image Analysis

- Patent Application No: 202441083307 (Approved)** – Pioneered privacy-preserving disease detection system combining federated edge learning with GANs to train AI models across 50+ healthcare facilities without centralizing patient data.

PROJECTS

FEELGAN - Privacy-Preserving Medical Image Analysis System (Patent Project)

Technologies: Python, TensorFlow, scikit-learn, NumPy, CNN, Flask

- Developed privacy-preserving disease detection system using Federated Edge Learning and Generative Adversarial Networks, improving diagnostic accuracy by **15%** across 50+ decentralized healthcare institutions
- Generated **10,000+ synthetic medical images** using GANs to augment limited datasets, improving model generalization across COVID-19, Tuberculosis, and OCT data achieving **90%+ accuracy** using TensorFlow and PyTorch
- Secured patent for FEELGAN system anticipated to streamline medical image analysis workflows across 50+ healthcare facilities

Full-Stack Real Estate Platform (MERN Stack)

Technologies: React, Node.js, Express, MongoDB, Socket.io, JWT, React Leaflet, HTML, CSS

- Designed modern, responsive UI with JWT authentication and Bcrypt hashing, reducing unauthorized access attempts by **40%** and enhancing overall data security
- Built full-stack features including user profile management, post creation, and real-time chat using Socket.io with interactive geolocation support via React Leaflet, improving user engagement by **60%**
- Implemented Context API for scalable and efficient state management, ensuring seamless user experience and reducing state-related bugs by **35%** during testing

AI-Powered Documentation Assistant with RAG

Technologies: Python, React JS, RAG, LLM

- Developed intelligent chatbot using Retrieval-Augmented Generation (RAG) with a large language model to provide context-aware Q&A from company documentation
- Preprocessed **1,000+ pages** of text by parsing, cleaning, and chunking content, implementing embeddings for better semantic search and retrieval
- Built scalable FastAPI backend with REST endpoints integrated with React frontend to support real-time question answering and efficient user interaction

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Express, Flask, FastAPI, TensorFlow, PyTorch, Keras, scikit-learn, OpenCV, LangChain

Databases & Tools: MongoDB, MySQL, Git, GitHub, Docker, Postman, Jupyter Notebook

AI/ML Specializations: Machine Learning, Deep Learning, Computer Vision, GANs, CNNs, Federated Learning, NLP, RAG, LLMs

ACHIEVEMENTS

- LeetCode** – Solved 350+ problems across data structures and algorithms
- Anveshan 2024** – Presented FEELGAN research at National Student Research Program
- SusHacks 2025** – Competed in Nationwide Innovation Hackathon